

Deep Research Task 3B: Known-Target Validation System Design

Executive Summary

The automated verification of American Sign Language (ASL) for targeted pedagogical applications represents a fundamentally distinct machine learning paradigm from generalized Continuous Sign Language Recognition (CSLR). In an educational application mirroring the Duolingo user experience, the system dictates the exercise and possesses a priori knowledge of the exact semantic target—whether a discrete lexical sign, a fingerspelled word, or a short phrase. Because the target is known and the temporal window is bounded (typically one to four seconds), the computational objective shifts entirely from open-vocabulary unconstrained decoding to known-target verification and diagnostic scoring. The architecture must not merely identify a sign; it must validate an attempt against a known class, measure its phonological fidelity compared to an expected idealized trajectory, and return actionable, sub-lexical feedback to the learner.

This report provides an exhaustive, implementation-ready specification for the known-target validation system. The analysis indicates that the optimal formulation for ASL pedagogical validation relies on a hybrid architecture: Prototypical Metric Learning utilizing an angular margin loss (such as ArcFace) to generate dense spatio-temporal embeddings, explicitly factorized into orthogonal phonological subspaces via Spatio-Temporal Graph Convolutional Network (ST-GCN) auxiliary heads. This formulation maps the ASL validation problem directly to the mathematical frameworks established by speech Computer-Assisted Pronunciation Training (CAPT) and Goodness-of-Pronunciation (GOP) algorithms. By replacing acoustic log-posteriors with spatial embedding cosine distances, the system can dynamically evaluate learner attempts against a dictionary of known signs, rejecting out-of-distribution motion while localizing errors to specific linguistic parameters, namely Handshape, Location, Movement, and Palm Orientation¹.

Furthermore, the design explicitly decouples isolated lexical validation from fingerspelling validation, acknowledging that routing between specialized heads is a product-level UI decision rather than a learned algorithmic router. For fingerspelled sequences, the system abandons fixed-vector metric embeddings in favor of Connectionist Temporal Classification (CTC) forced alignment. Because the target word is known in advance, the algorithm restricts the CTC decoding graph exclusively to the expected character sequence, yielding a precise frame-level temporal alignment and character-level confidence score that is highly resilient to the erratic coarticulation, hesitancy, and velocity shifts characteristic of novice signers¹. Through careful mitigation of the L1 (fluent) to L2 (novice) domain shift via canonical mirroring and temporal interpolation, this architecture establishes a robust foundation for real-time educational feedback.

Part 1: Problem Formulations to Compare

The architectural search space for ASL validation encompasses multiple formulations. Each carries distinct mathematical affordances and engineering constraints when applied to the specific domain of short bounded video windows, small active vocabularies, and learner-shifted kinematics. Prior to selecting the formulation, the foundational sensory apparatus—the landmark extractor—must be resolved based on evidence rather than inherited assumptions.

Resolution of Extractor Choice

The previous research phase presented a choice between MediaPipe Holistic and the RTMW/MMPose framework. The evidence strictly disqualifies MediaPipe Holistic as the primary inference engine for high-fidelity ASL validation. MediaPipe operates on a tracking-by-detection paradigm that frequently collapses during rapid motion blur and self-occlusion, resulting in severe temporal jitter¹. This jitter acts as adversarial noise to downstream temporal models, destroying the micro-kinematics necessary to distinguish minimal pairs (e.g., subtle wrist rotations). Conversely, RTMW achieves a state-of-the-art 70.2 mAP on the COCO-Wholebody benchmark while maintaining sub-15ms inference efficiency via ONNX runtime⁶. RTMW's coordinate classification mechanism reduces quantization errors, providing the occlusion robustness necessary for accurate sub-lexical parameter extraction. Therefore, the system will utilize RTMW as the primary deployment extractor, relegating MediaPipe strictly to a lightweight browser-fallback tier where local hardware cannot support ONNX execution⁶.

With the input tensor established as a normalized 65-node graph extracted via RTMW, the machine learning problem formulations are evaluated as follows.

Restricted Closed-Set Classification

The traditional approach treats sign recognition as a closed-set classification problem, terminating the network with a softmax layer representing the full trained vocabulary. To adapt this for verification, the system accepts the user's attempt if the known target corresponds to the top-1 prediction with a probability margin exceeding a predefined threshold. An explicit "garbage" or background class is trained to capture non-signing motion.

The expected accuracy of this approach is high on fluent, in-distribution data, mirroring the results of the Kaggle ISLR 1st Place solution which utilized a 1D Convolutional Neural Network (CNN) to achieve high accuracy within real-time latency budgets¹. However, the robustness degrades catastrophically on novice data. The data requirements are immense, as the model requires thousands of examples per class to prevent overfitting on the long-tail distribution⁷. More critically, the out-of-distribution (OOD) behavior of softmax is mathematically flawed for pedagogical verification. Softmax forces a probability distribution over known classes; if a learner scratches their nose, the network will confidently misclassify the movement as the closest known sign, leading to severe false acceptances⁸. Furthermore, global classification provides no mechanism for sub-lexical feedback. Engineering cost is low, but the architectural limitations render it unsuitable for the product requirements.

Binary Verification (One-vs-All Detectors)

Binary verification trains a distinct support vector machine, logistic regressor, or dedicated neural head for every single sign in the curriculum. The formulation asks: "Is this trajectory the sign for APPLE, or is it background noise?"

While this formulation simplifies the decision boundary for a specific sign, it is highly susceptible to false positives from minimal pairs unless the negative class is exhaustively curated. The data requirements necessitate extensive hard-negative mining per target. OOD behavior is moderate, as the detector can reject arbitrary motion if the negative class is well-sampled. However, this approach offers no native support for phonological feedback. The engineering cost is exceptionally high at scale; managing and tuning individual hyper-parameters, thresholds, and negative distributions for hundreds of distinct curriculum signs introduces unsustainable technical debt¹⁰.

Embedding + Metric Learning (Dictionary Retrieval)

This formulation reframes the task from classification to representation learning. The neural network projects the spatio-temporal sequence into a high-dimensional continuous latent space. Verification becomes a geometric distance calculation (e.g., cosine similarity) between the learner's generated embedding and the idealized reference embeddings (centroids) of the known target.

Evidence from the ASL Citizen dataset demonstrates the power of this approach, achieving a 91% recall-at-10 metric under strict signer-independent evaluation splits¹. By employing margin-based metric learning losses, such as ArcFace (Additive Angular Margin Loss), the network explicitly enforces intra-class compactness and maximizes inter-class variance¹¹. This mathematical separation is strictly required for differentiating phonological minimal pairs. The data requirements are low per class, enabling few-shot capability and allowing new curriculum signs to be added simply by storing new centroids, entirely bypassing network retraining¹³. The OOD behavior is excellent, as distance-based thresholding naturally rejects inputs that fall far from known cluster centroids. The engineering cost is moderate, requiring careful selection of triplet or contrastive mining strategies during training.

Template Alignment (Dynamic Time Warping)

Template alignment utilizes Dynamic Time Warping (DTW) or differentiable soft-DTW to measure the optimal alignment distance between the normalized pose trajectory of the learner and a stored reference trajectory of a fluent teacher.

This formulation is highly robust to temporal variations, explicitly addressing the L2 learner domain shift where novices sign much slower than native speakers¹⁶. The data requirements are negligible, requiring only a single pristine "gold standard" reference template. However, standard DTW can forcibly align garbage motion to the template, generating falsely optimistic alignment scores unless strict thresholding is applied to the total cumulative warp path cost. While neural-feature + DTW hybrids excel at temporal localization—mapping the exact frames of the learner's error to the teacher's timeline—the inference cost of executing DTW over deep feature tensors is computationally heavy and risks violating the strict 200-millisecond real-time budget¹⁷.

Forced-Alignment Scoring (CAPT/GOP-Style)

Adapted from spoken language Computer-Assisted Pronunciation Training (CAPT), this formulation utilizes a Goodness-of-Pronunciation (GOP) paradigm. A sequence-to-sequence model (such as a 1D-CNN with CTC) aligns the attempt to the target's expected phonological or character structure.

The GOP algorithm calculates the posterior probability of a target state given the observation, normalized against the probability of an unconstrained background loop². This formulation is exceptionally powerful for unsegmented, fluid, and highly coarticulated motion, making it the definitive standard for fingerspelling validation. It natively produces per-segment (or per-letter) scores, enabling precise temporal feedback²⁰. The engineering cost is moderate to high, requiring the integration of CTC loss or Hidden Markov Model (HMM) decoders into the inference graph⁴.

Formulation Comparison Matrix

Formulation	Expected Accuracy & Robustness	Data Requirements	Out-of-Distribution (OOD) Behavior	Feedback Support	Engineering Cost	Prior Art / Evidence Base
Softmax Classification	High on known fluent data; brittle on novice data.	High per-class requirements to prevent long-tail overfitting.	Poor. Vulnerable to overconfidence on unknown gestures ⁸ .	Minimal. Cannot isolate sub-lexical errors.	Low. Standard classification pipeline.	Kaggle ISLR 1st Place (1D CNN) ¹ .
Binary Verification	High for specific targets; struggles with minimal pairs.	Requires massive hard-negative mining per target.	Moderate. Rejects arbitrary motion if the negative class is well-sampled.	None natively.	High. Requires tuning distinct hyper-parameters for every sign.	Traditional keyword spotting paradigms ¹⁰ .

Embedding + Metric Learning (ArcFace)	Exceptional. Enforces margins crucial for distinguishing minimal pairs.	Low (Few-Shot capable). Scales without retraining ¹³ .	Excellent. Distance thresholding naturally rejects anomalies.	High. Can support semantic distance measurement and sub-lexical heads.	Moderate. Requires complex margin penalties and triplet mining.	ASL Citizen retrieval (91% R@10) ¹ ; PhonSSM ²³ .
Template Alignment (Soft-DTW)	Highly robust to temporal speed variations ¹⁶ .	Requires only a few pristine "gold standard" templates.	Moderate. DTW can forcibly align garbage motion without strict penalty tuning.	Exceptional for temporal localization.	High inference cost for deep tensors; borderline for real-time.	Speech CAPT alignment ²⁵ ; soft-DTW neural hybrids ¹⁷ .
Forced-Alignment Scoring (CTC/GOP)	Very high for unsegmented or coarticulated sequences.	Requires sequence-level alignments during training.	High. Log-likelihood ratio compares against an unconstrained loop ² .	Exceptional. Yields per-letter or per-segment scores natively.	Moderate. Requires integrating CTC decoders.	Speech MDD ²⁰ ; CTC ASL fingerspelling ⁴ .

Recommended Formulation and Fallback

The definitive formulation for the isolated lexical sign verifier relies on a **Hybrid Embedding + Metric Learning** architecture, specifically utilizing Prototypical Networks enhanced with an ArcFace angular margin loss, combined with orthogonal phonological subspaces.

The rationale is heavily grounded in the evidence surrounding the PhonSSM architecture and the ASL Citizen dataset. Softmax classification is entirely unsuited for pedagogical verification due to its inability to scale zero-shot to new vocabulary and its catastrophic failure in open-set

rejection¹. By mapping the spatio-temporal input sequence via a Spatio-Temporal Graph Convolutional Network (ST-GCN) into a high-dimensional latent space governed by an ArcFace loss, the system creates compact, distinct clusters for every sign¹¹. The verification decision becomes a highly calibrated cosine distance check against the known target centroid. Crucially, to satisfy the product requirement for actionable feedback, this global embedding space is explicitly factorized into four orthogonal sub-spaces corresponding to the phonological parameters of ASL: Handshape, Location, Movement, and Palm Orientation²³. The global embedding dictates the overarching accept/reject decision, while the auxiliary sub-lexical heads generate the diagnostic feedback.

If the orthogonal factorization of the latent space proves too computationally expensive for the edge hardware budget, the designated **fallback formulation** is a **Neural Feature + DTW Hybrid**. In this fallback, the ST-GCN extracts frame-level representations, and a highly optimized Dynamic Time Warping algorithm calculates the alignment distance against a stored teacher template¹⁰. Feedback is subsequently generated by identifying the specific temporal segment (e.g., the transitional movement) that incurred the highest localized warp cost.

Part 2: Decision Policy and Error Design

The pedagogical environment demands asymmetrical error handling. In the context of a language learning application, a false accept (verifying an incorrect sign as correct) is highly detrimental, as it reinforces improper muscle memory and phonological errors that will fossilize over time. Conversely, a false reject (failing a mathematically correct sign) frustrates the user, but this friction can be mitigated by product design, framing the rejection as an opportunity for micro-refinement rather than an absolute failure.

Target Operating Points and Calibration

The verification system cannot utilize a static, global threshold for acceptance. Signs possess radically different degrees of intra-class variance. The sign for "FATHER" has a highly stable and constrained spatial location (the thumb resting on the forehead), whereas directional verbs like "HELP" or "PAY" exhibit massive spatial variability depending on the object of the verb. Applying a global threshold across these disparate kinematic profiles ensures systemic failure¹.

Therefore, **per-sign thresholds** are mandatory. The system requires difficulty-dependent operating points that scale with the learner's progression. During the initial introduction of a new sign in the curriculum, the decision boundary should operate near the Equal Error Rate (EER) threshold, balancing leniency to prevent immediate user churn. As the learner advances to spaced-repetition review sessions, the threshold must dynamically tighten to prioritize a strictly low False Acceptance Rate ($FAR < 5\%$), enforcing precise phonological execution. The Expected Calibration Error (ECE) metric must be utilized during offline evaluation to ensure that the cosine similarity scores output by the ArcFace embedding space accurately reflect true empirical verification probabilities under learner-style (L2) input conditions⁷.

Unknown Rejection and Open-Set Recognition

A critical vulnerability of embedding spaces is their behavior when presented with data wholly outside the training distribution. To ensure that arbitrary motion (e.g., a user adjusting their

glasses or waving at the camera) cannot pass the verification check, the system must deploy robust Open-Set Recognition (OSR) techniques.

The **OpenMax algorithm** provides the optimal mathematical framework for this rejection. Unlike standard softmax, which normalizes outputs to sum to one regardless of the input's absolute proximity to the training data, OpenMax utilizes Extreme Value Theory (EVT)⁸. During the calibration phase, the algorithm fits a Weibull distribution to the distances between the training samples and their respective class centroids. During real-time inference, if a learner's generated embedding falls into the extreme tail of the target class's Weibull distribution—or sits in an unpopulated void of the latent space—the network's activation vector is forcefully suppressed. The system explicitly outputs a "Garbage/Unknown" state, signaling that the motion lacked the underlying syntactic structure of ASL⁹.

Confusable Mining and Minimal Pairs

A pervasive failure mode of kinematic extractors is the conflation of minimal pairs. Minimal pairs in ASL are signs that share three fundamental parameters but differ in only one, such as "WAKE" versus "AWAKE," which differ purely in the micro-tension of the movement¹. Standard distance thresholds frequently fail to resolve these pairs, as the overarching macro-trajectories are nearly identical.

The decision policy must explicitly incorporate confusable mining utilizing the ASL-LEX 2.0 phonological neighbor structure²³. For every target sign in the active curriculum, a static list of its top-K phonological neighbors is loaded into the decision policy dictionary. The validation rule demands a **margin-over-confusables**: the attempt's similarity to the target centroid must not only exceed the target's absolute acceptance threshold, but it must also exceed the

similarity to all defined minimal-pair centroids by a tuned margin Δ . If the margin is violated, the system triggers a targeted diagnostic rejection.

Decision Policy Pseudocode

The following Python-style pseudocode defines the exact sequence of validation, executing after the ST-GCN backbone has processed the $(B, T, 65, 10)$ input tensor.

Python

```
def validate_attempt(attempt_tensor, target_sign_id, curriculum_db, user_level):
```

```
    """
```

```
    Validates a known-target ASL attempt, providing hierarchical feedback.
```

Parameters:

attempt_tensor (Tensor): [65, 10] spatio-temporal features (root-centered, scaled).

target_sign_id (String): The dictionary ID of the expected sign.

curriculum_db (Dict): Pre-computed centroids, thresholds, and minimal pairs.

user_level (Float): Scales from 0.0 (lenient EER) to 1.0 (strict low FAR).

```
"""
# 1. Forward Pass: Extract global metric embedding and orthogonal sub-lexical logits
global_embed, phono_logits = model.forward(attempt_tensor)

# 2. Retrieve A Priori Target Parameters
target_data = curriculum_db[target_sign_id]
target_centroid = target_data.centroid
base_thresh_eer = target_data.eer_threshold
strict_thresh_far = target_data.low_far_threshold

# 3. Dynamic Threshold Interpolation based on pedagogical difficulty
active_thresh = base_thresh_eer + (user_level * (strict_thresh_far - base_thresh_eer))

# 4. Open-Set Garbage Rejection via Extreme Value Theory (OpenMax/Weibull)
dist_to_target = cosine_distance(global_embed, target_centroid)
weibull_inclusion_prob = calculate_weibull_probability(dist_to_target,
target_data.weibull_params)

if weibull_inclusion_prob < 0.05:
    return Reject(reason="Motion not recognized as signing. Please try again.")

# 5. Semantic Validation (Global Accept/Reject)
sim_score = cosine_similarity(global_embed, target_centroid)
if sim_score < active_thresh:
    # Failed global validation; generate specific sub-lexical feedback
    feedback = generate_phonological_feedback(phono_logits,
target_data.expected_phonemes)
    return Reject(reason="Inaccurate execution." + feedback)

# 6. Minimal Pair Discrimination (Margin-Over-Confusables)
margin_delta = 0.05 # Minimum required distance delta between the target and a near-miss
for confusable_id in target_data.minimal_pairs:
    conf_centroid = curriculum_db[confusable_id].centroid
    conf_sim = cosine_similarity(global_embed, conf_centroid)
    if (sim_score - conf_sim) < margin_delta:
        # The execution drifted too close to the phonological neighbor
        differential_feedback = isolate_minimal_pair_difference(target_sign_id,
confusable_id)
        return Reject(
            reason=f"Sign looked too much like '{confusable_id}'.",
            feedback=differential_feedback
        )

```

```
# 7. Verification Success
```

```
return capt(score=sim_score, feedback="Excellent execution!")
```

Part 3: Feedback Localization

The product differentiator for this system is the generation of actionable, localized feedback. Merely returning a binary rejection leaves the learner without a path to pedagogical correction. The feedback system must emulate a human tutor, leveraging the phonological decomposition of the sign to isolate exactly where the kinematic execution failed.

Sub-Lexical Parameter Scoring

Drawing heavily on the architectural breakthroughs of the Sem-Lex benchmark and the PhonSSM architecture, the spatial representation generated by the ST-GCN backbone is forced into orthogonal latent subspaces³. This factorization explicitly separates the intertwined visual features of the sign into distinct semantic channels. The model employs four auxiliary classification heads trained against the ASL-LEX 2.0 phoneme labels³:

1. **Handshape Head:** Classifies the finger and joint configuration into one of the fundamental ASL handshapes (e.g., '1' index, 'B' flat, '5' spread).
2. **Location Head:** Classifies the articulation point relative to the body into major regional clusters (e.g., forehead, chin, chest, neutral space).
3. **Movement Head:** Identifies the macro-trajectory of the sequence (e.g., linear, circular, repeated oscillatory motion).
4. **Palm Orientation Head:** Resolves the radioulnar rotation relative to the camera vector (e.g., palm facing in, out, up, down).

Empirical evidence from the Sem-Lex dataset demonstrates that a properly regularized ST-GCN can extract and classify these sub-lexical phonological parameters with an average accuracy of 85%¹. When a learner's global cosine similarity falls below the acceptance threshold, the verification system analyzes the output logits of these four heads. It compares the argmax prediction of the learner's physical execution against the hard-coded, expected phonemes of the target sign stored in the curriculum dictionary.

CAPT Feedback Granularity Evidence

Research in speech CAPT explicitly demonstrates that providing simultaneous, multi-parameter feedback induces severe cognitive overload, frustrating the learner and impeding progress¹⁸. If a novice executes the incorrect handshape, places it in the incorrect location, and utilizes the wrong movement trajectory, presenting all three errors simultaneously is counterproductive. The ASL feedback system must employ a **hierarchical blame assignment protocol**. The system ranks the phonological errors by their severity—calculated by measuring how far the sub-lexical logit fell below the expected class threshold—and returns strictly the single most egregious error. For example, the user interface will communicate: "Your location was correct, but check your handshape; ensure your thumb is tucked across the palm." Once the primary error is corrected in a subsequent attempt, the system may surface the secondary error if it

persists.

Temporal Localization

To isolate *when* the error occurred within the 1-to-4 second video window, the system calculates the temporal activation weights of the ST-GCN's cross-attention layers. Because the input tensor preserves the time dimension T , temporal pooling layers can map the maximum loss gradient back to specific frame clusters. If the error is isolated to the final 15 frames of a 60-frame sequence, the UI can visually highlight the terminus of the video playback timeline, indicating to the learner that they failed to hold the final requisite posture or truncated the movement trajectory prematurely.

Part 4: Learner Domain Shift

A fundamental trap in sign language machine learning is the assumption that models trained exclusively on fluent, native Deaf signers (L1) will generalize to hearing novices (L2). This domain shift is profound. Fluent signers compress their spatial trajectories to conserve energy, rapidly transition between concepts, and heavily coarticulate. Conversely, novices execute signs with much slower velocity, exaggerate the spatial envelope outside the standard bounds of the torso, and exhibit distinct hesitation pauses—known as movement epenthesis—between discrete actions¹. A model calibrated strictly on PopSign or ASL Citizen data will ruthlessly reject novices, assuming their hesitant kinematics represent entirely different semantic concepts.

Mitigations for Kinematic Shift

To counteract this domain shift, the verification model cannot rely on absolute temporal lengths or raw velocity thresholds to calculate the validation score. The system must implement the following mathematical mitigations:

1. **Temporal Interpolation (Speed Normalization):** The spatial trajectory must be entirely isolated from the user's execution speed. Following the architectural optimizations utilized by the Kaggle ISLR winners, the temporal dimension of the learner's sequence must be uniformly interpolated (e.g., via cubic spline) to a fixed sequence length of $T = 160$ frames prior to network ingestion, regardless of the actual physical execution time¹. This mathematically standardizes a 3.0-second hesitant attempt and a 0.5-second fluent execution into an identical temporal vector, preventing the model from penalizing novices purely for a slow cadence.
2. **Per-User Calibration:** After a learner successfully passes the first 20 signs in the curriculum, the system computes a rolling average of their spatial amplitude and velocity parameters. These user-specific scalars are utilized to subtly adjust the target centroids in the metric space, personalizing the decision boundaries to the user's specific motor idiosyncrasies and accommodating natural variations in body morphology.

Left-Handed Learners: The Canonical Mirroring Policy

Handedness presents a critical failure point in spatial representation. A left-handed signer performing a sign is essentially providing a mirrored 3D topological map to the network. Naive

horizontal-flip data augmentation during training attempts to teach the model to become invariant to handedness. However, this is linguistically destructive; ASL relies heavily on dominant versus non-dominant hand interactions, and spatial indexing from left-to-right carries specific semantic and syntactic meaning.

The evidence dictates an inference-time **canonical mirroring policy**, mirroring the preprocessing pipeline utilized in the creation of the PopSign dataset and recent ASL alignment models¹. The application must utilize an explicit user-profile toggle (or an initial automated heuristic based on wrist velocity) to declare dominant handedness. If the user is left-handed,

their extracted coordinate tensor (X, Y, Z) undergoes a deterministic horizontal reflection on the X-axis *before* entering the One Euro filter and normalization chain. Consequently, the neural network only ever models and evaluates right-handed canonical space. This completely eliminates the need to partition model capacity for left-handed topologies and guarantees that left-handed novices face the exact same algorithmic scrutiny as right-handed users¹.

Learner-Attempt Evaluation Set Plan

To validate the FAR/FRR thresholds safely and calibrate the OpenMax distributions prior to launch, a proprietary L2 dataset must be collected.

- **Scale:** 30 novice participants with zero prior ASL experience.
- **Protocol:** Participants attempt a 50-sign curriculum subset. They are shown a video of the sign and asked to replicate it three times. To generate targeted negative samples for minimal-pair evaluation, they are explicitly instructed to intentionally perform phonological near-misses (e.g., executing the sign for "APPLE" on the chin instead of the cheek).
- **Yield:** $30 \text{ users} \times 50 \text{ signs} \times 3 \text{ attempts} = 4,500$ true L2 attempts, ensuring statistically meaningful FAR/FRR curves and margin distributions against the fluent-trained centroids.

Part 5: Fingerspelling Validation Design (Forced Alignment)

Fingerspelling validation cannot inherit the metric learning architecture used for isolated lexical signs. Fingerspelling operates at a fundamentally different temporal frequency; standard ST-GCNs extracting macro-trajectories fail to resolve the rapid micro-occlusions of coarticulated fingerspelling. Furthermore, fingerspelling is a continuous transcription task, not an isolated classification task.

Because the pedagogical application knows exactly which word the learner is attempting to spell, generalized Sequence-to-Sequence (Seq2Seq) open decoding is completely unnecessary and computationally wasteful. The system will instead employ **CTC Forced Alignment**⁴.

The Forced Alignment Mechanism

In a standard Connectionist Temporal Classification (CTC) setup, the network predicts a

probability distribution over the 26 letters of the alphabet (plus a crucial 'blank' token) for every temporal frame t . Open decoding runs a beam search to find the most probable sequence of letters. In forced alignment, the path search is mathematically constrained to *only* evaluate paths that collapse into the known target word⁴.

1. **Input:** Dynamic RGB bounding-box crops of the dominant hand, combined with the 21-node dominant hand coordinates, pass through a dedicated 1D-CNN or Squeezeformer backbone¹.
2. **Alignment:** The CTC forward-backward algorithm calculates the probability of the restricted target path $P(\text{target} \mid X)$.
3. **Log-Likelihood Scoring:** This probability is converted into a Phoneme Log-Likelihood Ratio (PLLR) or Goodness of Pronunciation (GOP) score. The score of the forced path is compared against the score of the unconstrained best path (the denominator)².
4. **Actionable Feedback:** If the log-likelihood ratio drops significantly at frame t , the system identifies which letter was expected at that specific temporal juncture. The feedback UI outputs localized feedback: "Your spelling was clear until the letter 'E'. Make sure your fingers are fully contracted."

The expected gain of known-target forced alignment over open decoding is profound. Open decoding on novice fingerspelling suffers exceptionally high Character Error Rates (CER) because novices hesitate, inserting arbitrary blank frames or lingering on transitional handshapes. Forced alignment utilizes the known target to effectively bridge these coarticulation gaps, yielding highly calibrated confidence scores for pedagogical verification without requiring mathematically perfect transcription accuracy⁵. Prior benchmarks demonstrating jumps from ~67% accuracy in open decoding to >82% in forced alignment environments confirm this architectural necessity⁴.

Part 6: Evaluation Protocol

The evaluation protocol extends the Task 1 metrics to specifically quantify the pedagogical safety, rejection calibration, and feedback validity of the verifier. The evaluation splits must strictly adhere to signer-independence to prevent dataset leakage⁷.

1. **Per-Sign FAR/FRR/EER:** Global accuracy metrics are irrelevant for a curriculum app. The system must report the Equal Error Rate (EER) independently for all target signs, ensuring no single sign systematically rejects learners due to poor spatial normalization or overstrict thresholds⁷.
2. **Margin Distributions vs. Confusables:** The metric embedding space must be evaluated by plotting the cosine distance distribution of True Positives against the distance distribution of Intentional Minimal-Pair Negatives. The intersection area of these two distributions defines the theoretical limit of the system's sub-lexical discrimination.
3. **Expected Calibration Error (ECE) for L2 Input:** Ensures that if the system outputs an 85% confidence score on a learner attempt, the attempt is empirically correct exactly 85% of the time, mathematically validating the dynamic abstention and rejection

threshold mechanisms⁷.

4. **Feedback-Accuracy Metrics:** On a labeled error set (where human experts have tagged L2 attempts with specific phonological errors, e.g., "Wrong Handshape"), measure whether the auxiliary ST-GCN head correctly blames the exact parameter identified by the expert.
5. **Human-in-the-Loop Usefulness:** A qualitative UX protocol to determine if learners correct their posture faster and exhibit less session abandonment when provided with hierarchical phonological feedback versus a standard passive video-mirror loop.

Part 7: Experiment Roadmap

The path to an implementation-ready production model is divided into four priority-ordered stages, designed to isolate and resolve specific engineering risks consecutively.

- **Stage 1: Formulation Baseline vs. Prototype**
 - *Research Question:* Does a Prototypical ArcFace ST-GCN outperform the legacy MediaPipe Holistic + TCN + CTC prototype on isolated verification?
 - *Hypothesis:* The angular margin loss will cluster minimal pairs more effectively than the TCN's softmax, achieving higher recall at tighter false-accept thresholds.
 - *Success Criterion:* >90% True Accept Rate at a 5% False Accept Rate on a 50-sign fluent-signer test set.
 - *Failure Criterion:* The ArcFace embeddings collapse or show high spatial overlap between phonologically similar signs, indicating insufficient spatial resolution from the 65-node graph.
- **Stage 2: Learner Domain Shift Mitigation**
 - *Research Question:* Can temporal spline interpolation completely mitigate the velocity drop and movement epenthesis of L2 learners?
 - *Hypothesis:* Standardizing input length to $T = 160$ frames will equalize the feature variance between 0.5-second fluent clips and 3.0-second novice clips.
 - *Success Criterion:* Less than a 10% absolute drop in EER when testing the fluent-trained centroid model on the newly collected 30-participant L2 dataset.
 - *Failure Criterion:* The model mathematically rejects >30% of valid L2 attempts due to unique L2 spatial exaggerations that temporal interpolation cannot normalize.
- **Stage 3: Sub-Lexical Feedback Accuracy**
 - *Research Question:* Can ST-GCN auxiliary heads trained on ASL-LEX 2.0 labels accurately diagnose specific parameter failures in real-time?
 - *Hypothesis:* Extracting orthogonal subspaces from the latent embedding will yield >80% accuracy in predicting Handshape, Location, and Movement anomalies.
 - *Success Criterion:* The system correctly isolates the induced error (e.g., intentionally using the wrong location) on 85% of the L2 negative sample set.
 - *Failure Criterion:* The auxiliary heads over-correlate (e.g., a poor handshape execution mathematically suppresses the location score), rendering the hierarchical feedback system useless.
- **Stage 4: Open-Set Rejection Calibration**

- *Research Question*: Does OpenMax/Weibull modeling successfully reject arbitrary non-signing motion without accidentally rejecting valid but hesitant signs?
- *Hypothesis*: Modeling the extreme tails of the embedding clusters via EVT will clearly separate out-of-domain random motion from in-domain L2 hesitancy.
- *Success Criterion*: >95% True Rejection Rate of random idle/scratching gestures, with <2% impact on valid sign acceptance.
- *Failure Criterion*: OpenMax severely truncates the acceptance radius, causing a massive spike in False Rejects for valid users. (Fallback mechanism: Energy-based OOD detection).

Part 8: Open Questions

1. **3D Depth Ambiguity in Monocular Webcams**: While RTMW attempts to estimate a Z-axis, monocular depth extraction is notoriously unstable in consumer webcams. How severely will Z-axis jitter impact the Palm Orientation auxiliary head, given that palm direction relies entirely on the mathematical projections of 3D bone vectors?
2. **Threshold Calibration at Scale**: Tuning FAR/FRR thresholds for 100 signs manually is feasible, but as the curriculum scales to 1,000+ words, how can the system auto-calibrate optimal decision boundaries without requiring exhaustive, expensive L2 testing for every new vocabulary addition?
3. **Shoulder-Root Normalization Bias**: Normalizing the spatial coordinates by dividing by the absolute shoulder width ensures scale invariance. However, does this mathematically scrub the delicate spatial offsets required to distinguish location-based minimal pairs for users with atypical body morphologies (e.g., children or users in wheelchairs)?

Part 9: Handoff to Engineering

The following specifications construct the exact implementation requirements for the spatial architecture and data pipeline teams, finalizing the system design.

1. Input/Output Tensor Specification The stream buffer must output $(B, T, 65, 10)$

tensors to the network. The time dimension T is dynamically interpolated to a fixed sequence length of 160 frames prior to ingestion¹. The 65 nodes comprise 42 hand, 7 body, and 16 facial

coordinates. The 10 feature channels contain (x, y, z) root-centered scaled coordinates,

(v_x, v_y, v_z) instantaneous velocities, (b_x, b_y, b_z) directional bone vectors, and the

extractor confidence score c^6 . Missing upstream nodes with a confidence $c < 0.3$ must be explicitly zero-masked; the downstream ST-GCN must utilize learnable spatial imputation to prevent the processing of hallucinated RTMW joints⁶. The required model outputs are:

- **global_embedding**: A dense vector $\in \mathbb{R}^{512}$ trained via ArcFace margin loss.
- **phono_logits**: A tuple of four tensors representing Handshape, Location, Movement, and Orientation log-probabilities.

- `ctc_log_probs`: A vector $\in \mathbb{R}^{27}$ (Alphabet + Blank) for the dedicated fingerspelling CTC decoding path.

2. Decision Policy Parameters The data engineering team must generate a master `curriculum_db.json` defining the parameters for every sign. This dictionary must include the `centroid_vector`, `eer_threshold`, `low_far_threshold`, `weibull_params` (for OpenMax), and a definitive list of `minimal_pairs` cross-referenced from the ASL-LEX 2.0 phonological corpus²³. The architecture team must expose the `user_level` variable (0.0 to 1.0) to the product UI layer to allow dynamic tightening of the verification threshold as the learner progresses.

3. Training Data Mixture Primary representation learning and backbone training must utilize fluent L1 data from WLASL, ASL Citizen, and PopSign to define the idealized cluster centroids¹. Domain adaptation and fine-tuning must utilize the strictly isolated L2 dataset (collected in Stage 2) to calibrate the OpenMax Weibull distributions and tune the EER margins. Negative data mining must incorporate out-of-domain conversational gesturing, idle sitting, and face-touching to explicitly train the Open-Set Rejection boundaries³⁰.

4. Final Architecture Deep-Dive Requirements (Task 6) If the chosen ArcFace + ST-GCN formulation faces extreme latency constraints or memory bottlenecks during TensorFlow Lite or ONNX edge conversion, a final Task 6 architecture survey must exclusively evaluate the efficacy of replacing the ST-GCN temporal pooling layer with a linear-time State Space Model

(e.g., Mamba). This would theoretically achieve strictly $O(1)$ recurrent inference latency while preserving the dense 65-node spatial topology, though it introduces significant implementation risks outside standard deep learning toolchains.

Source Table

Source ID	Context / Topic	Relevance
6	Task 1 & 2 Specifications	Input tensor bounds (B, T, 65, 10), RTMW extractors, 3D normalization logic, and ECE evaluation metrics.
1	Task 3A Prior Art Evidence	Kaggle ISLR 1D CNNs, ASL Citizen dictionary retrieval (91% R@10), PopSign learner constraints, and left-handed canonical mirroring preprocessing.

2	Goodness of Pronunciation (GOP)	Defines log-likelihood ratio scoring for phonemes; provides the foundational mathematical basis for the verification decision policy and CAPT alignment.
4	CTC Forced Alignment	Details restricting decoding graphs to known character sequences, bridging coarticulation gaps natively for fingerspelling validation.
8	Open-Set Recognition & OpenMax	Utilizing Extreme Value Theory (Weibull distributions) to reject unknown gestures in visual processing and maintain closed-set performance.
23	ASL-LEX 2.0 / Minimal Pairs	Structure of phonological neighbors; defining orthogonal sub-lexical parameters (Handshape, Location, Movement) for margin-over-confusables rules.
13	Prototypical Networks	Baseline for mapping inputs to known centroids for few-shot metrics and scalable metric learning without retraining.
11	ArcFace Margin Loss	Additive angular margins to force intra-class compactness, critical for mathematically differentiating kinematic minimal pairs.

3	Sem-Lex / PhonSSM	Evidence demonstrating that ST-GCNs can isolate phonological features with 85% accuracy; provides the justification for auxiliary feedback heads.
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